

Sarvam-M Baseline Evaluation Report

1. Objective

This evaluation measures the performance of Sarvam’s chat model across English, Hindi, and Tamil using four task categories: Factual Question Answering, Translation, Summarization, and Reasoning/Math. The evaluation is intended to identify strengths, errors, language-specific behavior, and response efficiency.

2. Updated Evaluation Setup

The prompt matrix was updated to follow the assignment requirements: 100 prompts in total, with 25 prompts per task.

Task	Prompts	Required source	Evaluation method
Factual QA	25	Hand-written or IndicQA	Exact/reference-based correctness
Translation	25	FLORES-200 devtest; En↔Hi and En↔Tamil	chrF / chrF++ using SacreBLEU
Summarization	25	Short news paragraphs	Human faithfulness and fluency, 1–5
Reasoning/Math	25	GSM8K sample; some translated to Hindi	Final-answer exact match

The API evaluation uses temperature = 0.0 for reproducibility. The completed live run used the currently supported sarvam-105b model because the originally specified sarvam-m model is no longer available through the API. The pipeline records model outputs, latency, token usage, status, and errors in results.csv.

3. Dataset Update Status

The prompt matrix was revised today so that the translation and reasoning sections are aligned with the assignment's required benchmark structure. Rows 1–25 remain factual QA, rows 51–75 remain summarization, rows 26–50 form the translation block, and rows 76–100 form the reasoning/math block.

Evaluation-control note: the previously completed API run was executed against the earlier prompt set. Its numerical scores therefore must not be presented as scores for the newly revised prompts until the evaluation is rerun. This keeps every score traceable to the exact prompt row used to generate it.

4. Results From the Previously Completed Run

These results belong to the earlier completed 100-prompt run and are retained as a baseline reference only. They should be replaced after rerunning the API against the revised prompt matrix.

Task	Prompts	Previous result
Factual QA	25	21/25 correct (84%)
Translation	25	14.09 average chrF
Summarization	25	4.45/5 across 21 rated outputs
Reasoning/Math	25	23/25 correct (92%)

5. Previous Run — Language-Level Results

For the previous run, Factual QA achieved 90% for English, 80% for Hindi, and 80% for Tamil. Reasoning/Math achieved 90% for English, 90% for Hindi, and 100% for Tamil. Translation averages were 15.48 chrF for English-source prompts, 5.91 for Hindi-source prompts, and 19.28 for Tamil-source prompts. These values describe the previous prompt set and are not new results for the revised dataset.

6. Observed Failure Patterns From the Previous Run

The factual evaluation included cases where answers were semantically correct but did not exactly match the reference representation, such as “100 °C” versus “100 degrees Celsius”, English planet names in Indic-language answers, and transliterated Tamil forms. These examples show the sensitivity of strict reference matching to representation and script.

The previous translation evaluation produced a low average chrF on its hand-created prompt set. Because the translation block has now been revised toward the benchmark requirements, the translation metric should be recomputed from the revised prompts before drawing conclusions about translation quality.

7. Reproducibility and Traceability

- Model: sarvam-105b
- Temperature: 0.0
- Total evaluation size: 100 prompts
- Required structure: 25 prompts per task
- results.csv records model output, latency, token counts, status, and errors
- scores.csv records row-level scores traceable to prompt IDs
- The final report should use scores generated from the same revised prompts.csv used for the final API run

8. Conclusion

The baseline evaluation pipeline is operational and has successfully completed a 100-request run using sarvam-105b at temperature 0.0. The benchmark prompt matrix has now been revised toward the assignment requirements. The earlier numerical results remain useful as a baseline, but the final report must be regenerated after rerunning all 100 revised prompts so that scores, failure examples, latency, and token usage correspond to the final dataset.

Final submission note: Do not report the previous scores as the final scores for the revised benchmark. Run the evaluation again and regenerate results.csv and scores.csv first.